

L.P. Optimization Methods: Chapter 2 Linear Programming

1. Standard Formulation of Linear Programming

Objective Function: $\min C^T x \Leftrightarrow \sum_{j=1}^n C_j x_j$

Constraint subject to s.t. $Ax = b$
 $x \geq 0$
 $\sum_{j=1}^n A_{ij} x_j \leq b_i \quad i=1, \dots, m \quad x_j \geq 0 \quad j=1, \dots, n$

Decision Variables: $C: [R^n] \quad A: [R^{m \times n}] \quad x: [R^n] \quad b: [R^m]$

if meet constraint ~~look like~~ "inequality" $Ax \leq b$

then use slack variables: $\min \sum_{j=1}^n C_j x_j + \sum_{i=1}^m 0 \cdot x_{n+i}$

$\min C^T x + 0^T x_s$
s.t. $Ax + Ix_s = b$
 $x \geq 0$
 $x_s \geq 0$

$\Leftrightarrow$

s.t. $\sum_{j=1}^n A_{ij} x_j + x_{n+i} = b_i \quad i=1, \dots, m$
 $x_j \geq 0 \quad j=1, \dots, n$
 $x_{n+i} \geq 0 \quad i=1, \dots, m$

if max object function / s.t. $Ax > b$ / $x_k \leq 0$ Then multiply "-1"

if No constraint on x_k , then let $x_k = x_k^+ - x_k^-$
 $x_k^+, x_k^- \geq 0$

non object function

equality constraints

non-negative decision variables.

2. Optimal condition of Linear Programming

standard formulation: $\min C^T X$ $\Rightarrow$ $\min C_B^T X_B + C_N^T X_N$
 s.t. $AX = b$ $\Rightarrow$ s.t. $BX_B + NX_N = b$
 $X \geq 0$ $X_B \geq 0$ $X_N \geq 0$

$$A \begin{bmatrix} B & N \end{bmatrix} \begin{matrix} \text{max} \\ \text{invertible} \end{matrix} \begin{matrix} \text{min} \\ \text{max} \end{matrix}$$

From $BX_B + NX_N = b$
 and B is invertible

Therefore $X_B = B^{-1}b - B^{-1}NX_N$

m. X_B Basic variable & n-m X_N non-Basic Variable.

then the object function becomes: $C_B^T B^{-1}b + \underbrace{(C_N^T - C_B^T B^{-1}N)}_{\text{reduced cost}} X_N$

Because $X_N \geq 0$, if $(C_N^T - C_B^T B^{-1}N) \geq 0$ then the minimum value of object function will be $C_B^T B^{-1}b$

$X_N = 0$ minimizes the

Moreover, the optimal solution is $X_B = B^{-1}b$ & $X_N = 0$, so $B^{-1}b \geq 0$

if the objective function is max $\sim$. then $(C_N^T - C_B^T B^{-1}N)$ should ≤ 0

In conclusion, the optimal condition turns out to be:

"if there exists an invertible ^{sub} matrix B of matrix A such that $A = (B, N)$ which makes the linear programming problem can be reformulated as:

$$\min C_B^T X_B + C_N^T X_N$$

$$\text{s.t. } BX_B + NX_N = b$$

$$X_B \geq 0 \quad X_N \geq 0$$

with $X^T = (X_B^T, X_N^T)$ and the matrix B satisfied $B^{-1}b \geq 0$ &
 $C^T = (C_B^T, C_N^T) (C_N^T - C_B^T B^{-1}N) \geq 0$

Then optimal solution is: $X_B^* = B^{-1}b$, $X_N^* = 0$; optimal value of o.f. is $C_B^T B^{-1}b$

3. Duality of Linear Programming

standard formulation: $\min C^T x$
s.t. $Ax=b$
 $x \geq 0$ $\leftarrow$ primal problem

Suppose that x^* is the optimal solution of the primal problem
Define a function. Substitute x^* ^{into} the formulation

$$q(\lambda) = \min_{x \geq 0} \{C^T x + \lambda^T (b - Ax)\} \leq \underbrace{C^T x^*}_{\text{optimal value of primal problem}} + \underbrace{\lambda^T (b - Ax^*)}_0$$

because of "it is minimum" so it must at least lower than

Therefore.

$$\max_{\lambda} q(\lambda) \leq \min_{x \geq 0} C^T x \quad \{x | Ax=b, x \geq 0\}$$

tightest ~~the~~ lower bound of the primal problem

$$q(\lambda) = \min_{x \geq 0} \{C^T x + \lambda^T (b - Ax)\} = \lambda^T b + \underbrace{(C^T - \lambda^T A) x}_{\min_{x \geq 0} \begin{cases} 0 & \text{if } C^T - \lambda^T A \geq 0 \\ -\infty & \text{otherwise} \end{cases}}$$

$$\text{so } q(\lambda) = \begin{cases} \lambda^T b, & \text{if } C^T - \lambda^T A \geq 0 \\ -\infty, & \text{otherwise} \end{cases}$$

The Primal Problem $\iff$ The Dual Problem

$$\begin{array}{ccc} \min C^T x & \lambda^T = C^T B^{-1} & \max \lambda^T b \\ \text{s.t. } Ax=b & \iff & \text{s.t. } \lambda^T A \leq C^T \\ x \geq 0 & & \end{array}$$

No constraints on λ

$\lambda^T b \leq C^T x$ "weak duality" duality gap: $C^T x - \lambda^T b$

$u = u^*$ "strong" case: $u \leq -\infty$ & $+\infty \leq u \Rightarrow$ no feasible solution.

Primal Problem

ex. $\max C^T X \Rightarrow$ Define $q(\lambda)$ function as

$$\text{s.t. } AX \leq b \\ X \geq 0$$

$$q(\lambda) = \max_{X \geq 0} \{ C^T X + \lambda^T (b - AX) \}$$

(not standard form)

Suppose X^* is the optimal solution of primal problem.

$$q(\lambda) = \lambda^T b + (C^T - \lambda^T A) X \quad \text{then if } C^T - \lambda^T A \leq 0 \\ X \geq 0$$

then $X=0$ will maximize the $q(\lambda)$ will be $\lambda^T b \geq C^T X^* + \lambda^T (b - AX^*)$

Substitute X^* into $q(\lambda) \geq 0$

$$\geq C^T X^* = \max C^T X \\ (\forall \lambda \geq 0)$$

$$\therefore \min_{\lambda} q(\lambda) \geq \max C^T X.$$

($\lambda \geq 0$)

$\therefore$ Therefore, the dual problem: $\min \lambda^T b$

$$\text{s.t. } \lambda^T A \geq C^T$$

$\lambda \geq 0$

4. Simplex Method & Dual Simplex Method.

Optimality condition of L.P: $C_N^T - C_B^T B^{-1} N \geq 0$ & $B^{-1} b \geq 0$

Optimal value: $C_B^T B^{-1} b$

	C_B	C_N	
	X_B	X_N	
C_B	B	N	b

$\Rightarrow$

	C_B	C_N	
	X_B	X_N	
C_B	I	$B^{-1} N$	$B^{-1} b$
	0	$C_N^T - C_B^T B^{-1} N$	$C_B^T B^{-1} b$

Solving the primal and the dual problem alternatively!

$$\begin{array}{c}
 \begin{array}{ccc}
 I & B^{-1}N & B^{-1}b \\
 \left[\begin{array}{ccc} 1 & 0 & \dots \\ 0 & 1 & \dots \\ \vdots & \vdots & \ddots \end{array} \right] & \left[\begin{array}{ccc} -B^{-1}N_{1j} & B^{-1}N_{1k} & B^{-1}N_{1j} \\ B^{-1}N_{ik} & B^{-1}N_{ij} & B^{-1}N_{ij} \\ \vdots & \vdots & \vdots \end{array} \right] & \left[\begin{array}{c} (B^{-1}b_i - B^{-1}N_{ij} \frac{B^{-1}b_i}{B^{-1}N_{ij}}) \\ B^{-1}b_i \\ \vdots \end{array} \right] \\
 m \times m & m \times (n-m) & m \times 1 \\
 \left[0 \dots 0 \right] & \left[\dots C_N - C_B B^{-1}N \dots \right] & \\
 & < 0 & \\
 & \downarrow \text{most negative column } j \Rightarrow \text{entering basic variable} &
 \end{array}
 \end{array}$$

leaving basic variable $i = \arg \min \left\{ \frac{B^{-1}b_i}{B^{-1}N_{ij}}, \forall i \text{ with } B^{-1}N_{ij} \geq 0 \right\}$

Use Two-stage method to find the initial basic feasible solution: (for LPO)

LPO: $\min C_1x_1 + \dots + C_nx_n$

s.t. $a_{11}x_1 + \dots + a_{1n}x_n = b_1$

$a_{m1}x_1 + \dots + a_{mn}x_n = b_m$

$x_1, \dots, x_n \geq 0$

$$\begin{bmatrix} x_{n+1} \\ \vdots \\ x_{n+m} \end{bmatrix} = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}$$

LPI: $\min x_{n+1} + \dots + x_{n+m}$

s.t. $a_{11}x_1 + \dots + a_{1n}x_n + x_{n+1} = b_1$

the optimal solution for LPI is

$= b_1$

$a_{m1}x_1 + \dots + a_{mn}x_n + x_{n+m} = b_m$

$x_1, \dots, x_n \text{ \& } x_{n+1}, \dots, x_{n+m} \geq 0$

e.x. min $f(x_1, x_2) = -2x_1 - 3x_2$

s.t. $x_1 + 2x_2 \leq 8$

$4x_1 \leq 16$

$4x_2 \leq 12$

$x_1, x_2 \geq 0$

rewrite to its
Standard form

$\Rightarrow$

min $-2x_1 - 3x_2 + 0x_3 + 0x_4 + 0x_5$

s.t. $x_1 + 2x_2 + x_3 = 8$

$4x_1 + x_4 = 16$

$4x_2 + x_5 = 12$

$x_1, x_2 \geq 0, x_3, x_4, x_5 \geq 0$

Basic feasible solution $(0, 0, 8, 16, 12)$

objective function value: 0

	x_1	x_2	x_3	x_4	x_5	b	
$0x_3$	1	2	1	0	0	8	$\frac{8}{2} = 4$
$0x_4$	4	0	0	1	0	16	$\frac{16}{4} = 4$
$0x_5$	0	4	0	0	1	12	$\frac{12}{4} = 3$
	-2	-3	0	0	0	0	

enter x_2
leave x_5

$0x_3$	1	0	1	0	$-\frac{1}{2}$	2	$\frac{2}{1} = 2$
$0x_4$	4	0	0	1	2	16	$\frac{16}{4} = 4$
$-3x_2$	0	1	0	0	$\frac{1}{4}$	3	$\frac{3}{1} = 3$

$C_B^T B^{-1} b = 0 \cdot 2 + 0 \cdot 16 + (-3) \cdot 3 = -9$

$-2x_1$	1	0	1	0	$-\frac{1}{2}$	2	$\frac{2}{1} = 2$
$0x_4$	0	0	-4	1	$\frac{1}{2}$	8	$\frac{8}{2} = 4$
$-3x_2$	0	1	0	0	$\frac{1}{4}$	3	$\frac{3}{0.15} = 12$

enter x_1
leave x_3

enter x_5
leave x_4

$-2x_1$	1	0	0	$\frac{1}{4}$	$-\frac{1}{4}$	4
---------	---	---	---	---------------	----------------	---

Optimal solution: $(4, 2, 0, 0, 4)$

$0x_5$	0	0	-2	$\frac{1}{2}$	1	4
--------	---	---	----	---------------	---	---

$-3x_2$	0	1	0	$\frac{1}{2}$	0	2
	0	0	$\frac{3}{2}$	$\frac{1}{8}$	0	-8-6 = -14

Optimal value

Dual Simplex method:

$$C^T - C_B^T B^{-1} N \geq 0$$

I.

$$B^{-1} N$$

$$B^{-1} b$$

I.		$B^{-1} N$		$B^{-1} b$
l	$1 \quad \dots \quad 0$	$\frac{B^{-1} N_{lk}}{B^{-1} N_{ij}}$	$B^{-1} N_{lj}$	$B^{-1} b_l$
i	$0 \quad \dots \quad 1$	$\frac{B^{-1} N_{ik}}{B^{-1} N_{ij}}$	$B^{-1} N_{ij}$	$B^{-1} b_i$
$m \times m$		$m \times (n-m)$		m

$B^{-1} b_i < 0$
infeasible solution

$\Downarrow$
i-leaving basic variable

$$[0 \quad \dots \quad 0] \quad [C^T - C_B^T B^{-1} N] \quad \uparrow \quad j$$

entering basic variable

$$(C^T - C_B^T B^{-1} N)_k - \frac{B^{-1} N_{ik}}{B^{-1} N_{ij}} (C^T - C_B^T B^{-1} N)_j \geq 0$$

$j = \arg \min_k \left\{ \frac{(C^T - C_B^T B^{-1} N)_k}{|B^{-1} N_{ik}|}, \forall k \text{ with } (B^{-1} N)_{ik} < 0 \right\}$

MILP:

5. Standard Formulation of Mixed Integer Linear Programming

$$\begin{aligned} \text{MILP: } \min & C^T x + d^T y \\ \text{s.t. } & Ax + By = b \\ & x, y \geq 0 \\ & x \in \mathbb{R}^n, y \in \mathbb{Z}^l \end{aligned}$$

$$\begin{aligned} \text{ILP: } \min & C^T x \\ \text{s.t. } & Ax = b \\ & x \geq 0 \\ & x \in \mathbb{Z}^n \end{aligned}$$

* Branch and Bound Algorithm:
"B&B"

Branching
Bounding
Truncating
Backtracking

$$\begin{aligned} \text{ex. ILP: } \min & -40x_1 - 90x_2 \\ \text{s.t. } & 9x_1 + 7x_2 \leq 56 \\ & 7x_1 + 20x_2 \leq 70 \end{aligned}$$

$$x_1, x_2 \in \mathbb{Z}^+ \cup \{0\}$$

relaxed first

$$\min -40x_1 - 90x_2 + 0x_3 + 0x_4$$

$$\text{s.t. } 9x_1 + 7x_2 + x_3 = 56$$

$$7x_1 + 20x_2 + x_4 = 70$$

$$x_1, x_2, x_3, x_4 \geq 0$$

$$\begin{array}{c|cccc|c} & -40 & -90 & 0 & 0 & \\ \hline 0x_3 & 6.55 & 70 & 1 & 0 & 56 \\ 0x_4 & 7 & 20 & 0 & 1 & 70 \end{array}$$

$$\begin{array}{c|cccc|c} & -40 & -90 & 0 & 0 & \\ \hline X_3 & 6.55 & 0 & \frac{20}{131} & -\frac{7}{20 \cdot 131} & 8.55 \\ X_2 & \frac{7}{20} & 1 & 0 & \frac{1}{20} & 3.5 \end{array}$$

$$\begin{array}{c} -40x_1 \quad 1 \quad 0 \quad - \quad - \\ -90x_2 \quad 0 \quad 1 \quad - \quad - \end{array}$$

$$\therefore C_B^T B^{-1} b = -356.2 = z$$

$$\begin{aligned} 1^{\circ} \quad x_1 &= 4 \\ x_2 &= 2.1 \\ z &= -349 \end{aligned}$$

$$\begin{aligned} x_1 &\geq 5 \\ x_1 &= 5 \\ x_2 &= 1.57 \\ z &= -341 < -340 \text{ or} \end{aligned}$$

$$\begin{aligned} 2^{\circ} \quad x_2 &\leq 2 \\ x_1 &= 4 \\ x_2 &= 2 \\ 3^{\circ} \quad x_2 &\geq 3 \\ x_1 &= 1.42 \\ x_2 &= 3 \end{aligned}$$

$$\begin{aligned} x_2 &\leq 1 \\ x_1 &= 5.44 \\ x_2 &= 1 \\ z^* &= 308 \text{ Truncate} \end{aligned}$$

Truncate here
No feasible solution

$$(-340) \text{ first bound } z = -326.8 > -340 \times \therefore \text{Truncate}$$

$$1^{\circ} (x_1=4) \quad -160 - 90x_2$$

$$\min -90x_2$$

$$\text{s.t. } x_2 \leq \frac{20}{7} = 2.857$$

$$x_2 \leq \frac{21}{10} = 2.1$$

$$x_2 \geq 0 \quad \Downarrow$$

$$0 \leq x_2 \leq 2.1 \Rightarrow x_2 = 2.1$$

$$Z = -160 - 90 \cdot \frac{21}{10} = -349$$

$$2^{\circ} x_2 = 2$$

$$\min -40x_1$$

$$\text{s.t. } \begin{cases} 9x_1 + 14 \leq 56 & x_1 \leq 4.667 \\ 7x_1 + 40 \leq 70 & x_1 \leq 2.857 \\ x_1 \geq 0 \end{cases} \quad \& \quad x_1 \leq 4$$

$$\Rightarrow x_1 = 4$$

$$3^{\circ} x_2 = 3 \quad x_1 \leq 4$$

$$\text{s.t. } \begin{cases} 9x_1 + 21 \leq 56 & x_1 \leq 3.88 \\ 7x_1 + 60 \leq 70 & x_1 \leq 1.42 \\ x_1 \geq 0 \end{cases} \Rightarrow x_1 = 1.42$$

$\therefore$ Optimal solution

$$x_1^* = 4, x_2^* = 2$$

$$Z^* = -340$$

Cutting-Plane Algorithm.

$$\min C^T x$$

$$\text{s.t. } Ax = b$$

$$x \geq 0$$

First step 1
Solve relaxed L.P.

$$\text{Step 2: } Ax = b$$

$$\Downarrow$$

$$XB + B^{-1}N x_N = B^{-1}b$$

ith row:

$$(x_B)_i + \sum_{j \in N} [B^{-1}N]_{ij} (x_N)_j \leq (x_B)_i + (B^{-1}b)_i = (B^{-1}b)_i$$

$\otimes$ extra
Add inequality constraint.

$$(x_B)_i + \sum_{j \in N} [B^{-1}N]_{ij} (x_N)_j \leq (B^{-1}b)_i$$

if $(x_B^*)_i = (B^{-1}b)_i$ is $\hat{a}$ fractional
can not satisfy

	x_1	x_2	x_3	x_4	x_5	x_6
$40x_1$	1	0	0	0	1	0
$90x_2$	0	1	0	$\frac{401}{20}$	$\frac{7}{20}$	2.1
$0x_3$	0	0	1	0	$\frac{-131}{20}$	5.3
x_4	0	0	0	$\frac{1}{2}$	$\frac{1}{2}$	0
				$\frac{4.5}{121}$	$\frac{18.5}{121}$	
				90	1.2917	

$z = 0.15$
 dual
 leaving x_6
 entering x_5

non-integer solution X

necessary to ensure the slack variables in the set of basic variables be integers!

$$x_2 + x_6 = 2$$

$\Downarrow$

$x_1^* = 4, x_2^* = 2, z^* = 340$
 optimal solution & value

ex. $\max -40x_1 - 90x_2$

s.t. $9x_1 + 7x_2 \leq 56$

$7x_1 + 20x_2 \leq 70$

$x_1, x_2 \in \mathbb{Z}^+ \cup \{0\}$

Standard form relaxed L.P. had been solved as:

	-40	-90	0	0	
	x_1	x_2	x_3	x_4	
$-40x_1$	1	0	$\frac{20}{131}$	0	$\frac{630}{131} \approx 4.81$
$-90x_2$	0	1	$\frac{7}{131}$	$\frac{1}{20}$	$\frac{238}{131} \approx 1.82$
	0	0	$\frac{170}{131}$	$\frac{9}{2}$	$z^* = -355.88$

Since x_1^* is a fractional instead of an integer

Therefore.

the first row $x_1 + \frac{20}{131}x_3 = \frac{630}{131}$

$\Downarrow$ becomes new extra constraint

x_1	x_2	x_3	x_4	x_5
1	0	$\frac{20}{131}$	0	4.81
0	1	$\frac{7}{131}$	$\frac{1}{20}$	1.82
0	0	$\frac{170}{131}$	$\frac{9}{2}$	10001

$[1] \quad x_1 + [0.1526] x_3 \leq [4.81]$

$[1] \quad x_1 + [0.1526] x_3 \leq [4.81]$

$\therefore x_1 \leq 4 \Rightarrow x_1 + x_5 = 4$
 $x_5 \geq 0$

x_1	x_2	x_3	x_4	x_5	b
1	0	$\frac{20}{131}$	0	0	$\frac{630}{131} \quad 4$
0	1	$\frac{7}{131}$	$\frac{1}{20}$	0	$\frac{238}{131} \quad 1.82$
0	0	$\frac{170}{131}$	$\frac{9}{2}$	0	$\frac{106}{131} \quad 0.81$

use dual Simplex method!
 leaving x_5
 enter x_3

$z^* = -40 \cdot 4 - 90 \cdot \frac{2}{131} = -349$

But the x_2 is a fractional: 2.1

Therefore the second row becomes: $x_2 + \frac{1}{20}x_4 + \frac{7}{20}x_5 \leq 2.1$

namely $x_2 - x_5 \leq 2$

$x_6 \geq 0, \quad x_2 - x_5 + x_6 = 2$

NLP

6. Optimality condition for Non-linear Programming

$\min f(x)$ $\rightarrow$ Objective function
 subject to $g(x) \leq 0$ $\rightarrow$ Constraint function
 function are not linear in x
 $\mathcal{X} = \{x \mid g(x) \leq 0\}$

for unconstrained NLP:

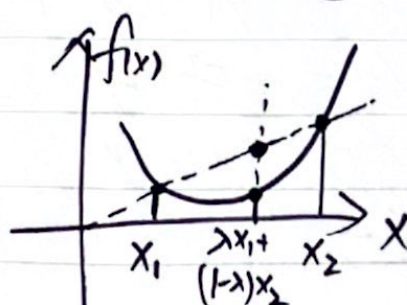
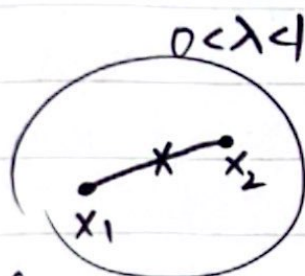
$$\nabla f(x^*) = 0 \quad \& \quad \nabla^2 f(x^*) \succeq 0$$

$\left(\frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_n} \right)^T$
 Hessian matrix of $f(x)$
 $[\dots]_{n \times n}$

> 7. Convex Set/Function/Programming

$\downarrow$
 For $\forall x_1, x_2 \in X$
 $\lambda x_1 + (1-\lambda)x_2 \in X$ holds

For $\forall x_1, x_2 \in X$
 $\rightarrow f[\lambda x_1 + (1-\lambda)x_2] \leq \lambda f(x_1) + (1-\lambda)f(x_2)$
 (< strictly) holds



* Definition Convex Programming: For a nonlinear programming

$$\min f(x) \quad \text{s.t. } x \in X$$

$\downarrow$
 if convex function & $\downarrow$ convex set, then this NLP is "Convex Programming"

1° local minimizer is its global minimizer

2° if f is strict convex function & X is a closed convex set $\Rightarrow$ Unique

Campus

3° if f is continuous & differentiable, then $\nabla f(x^*)^T (x - x^*) \geq 0 \quad \forall x \in X$ be a global minimizer of f over X
 moreover if X also a open set, then $\nabla f(x^*) = 0 \Leftrightarrow$

Karush-Kuhn-Tucker

for constrained NLP:

KKT condition for a NLP with inequality constraints

$$\begin{aligned} \min f(x) \\ \text{s.t. } g_j(x) \leq 0 \quad j=1, \dots, m. \end{aligned}$$

the necessary condition for x^* to be a local minimizer is:

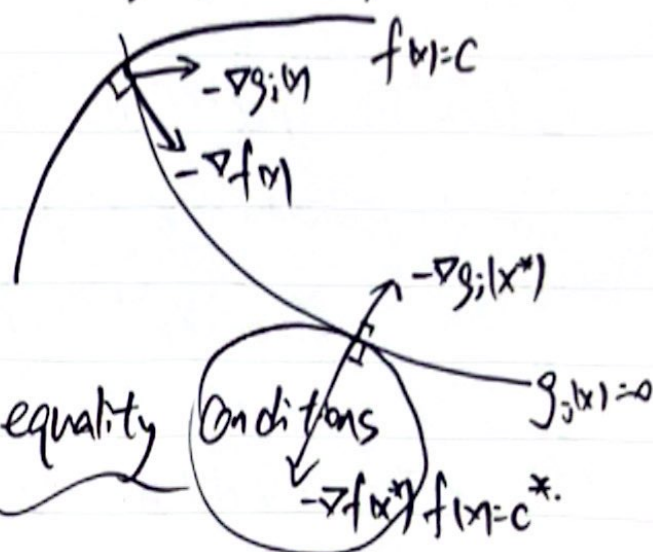
There exists a group of parameters λ_j^* , $j=1, \dots, m$. Such that
Lagrange multipliers

$$\nabla f(x^*) + \sum_{j=1}^m \lambda_j^* \nabla g_j(x^*) = 0$$

$$\textcircled{1} \begin{cases} \lambda_j^* \nabla g_j(x^*) = 0, j=1, \dots, m \\ \lambda_j^* \geq 0, j=1, \dots, m \end{cases}$$

if $\lambda_j = 0$ then $\lambda_j \geq 0$
if $g_j(x) < 0$ then $\lambda_j = 0$
min $f(x)$

with equality

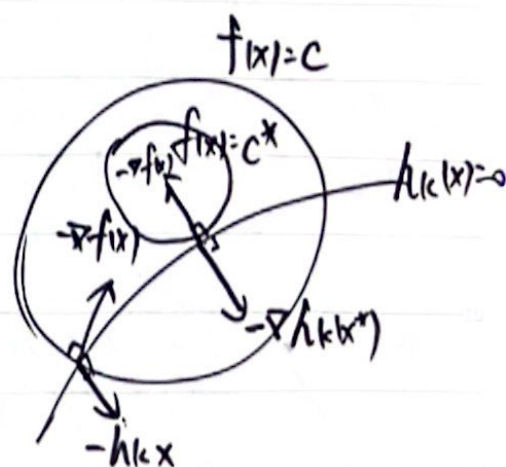


$$\text{s.t. } h_k(x) = 0, k=1, \dots, L$$

exists group of μ_k^* , $k=1, \dots, L$

$$\text{Such that } \nabla f(x^*) + \sum_{k=1}^L \mu_k^* \nabla h_k(x^*) = 0$$

$$\textcircled{2} \begin{cases} h_k(x^*) = 0, k=1, \dots, L. \end{cases}$$



$$L(x, \mu) = f(x) + \mu^T h(x) = f(x) + \sum_{k=1}^L \mu_k h_k(x) \begin{cases} \nabla_x L(x^*, \mu^*) = 0 \\ \nabla_\mu L(x^*, \mu^*) = 0 \end{cases}$$

with both inequality & equality

add $\textcircled{1}$ & $\textcircled{2}$

e.g. $\min_{x_1, x_2} f(x_1, x_2)$
 s.t. $x_1^2 + x_2^2 = 2$ $h(x_1, x_2) = x_1^2 + x_2^2 - 2$

$$L(x, \mu) = f(x_1, x_2) + \mu h(x_1, x_2)$$

$$\begin{cases} \nabla_x L(x, \mu) = (1 + 2\mu x_1, 1 + 2\mu x_2) = 0 \\ \nabla_\mu L(x, \mu) = x_1^2 + x_2^2 - 2 = 0 \end{cases}$$

$$\therefore \mu = \frac{1}{2}, (x_1^*, x_2^*) = (-1, -1) \quad f(x_1^*, x_2^*) = -2$$

8. Iterative procedure to approach the optimal solution

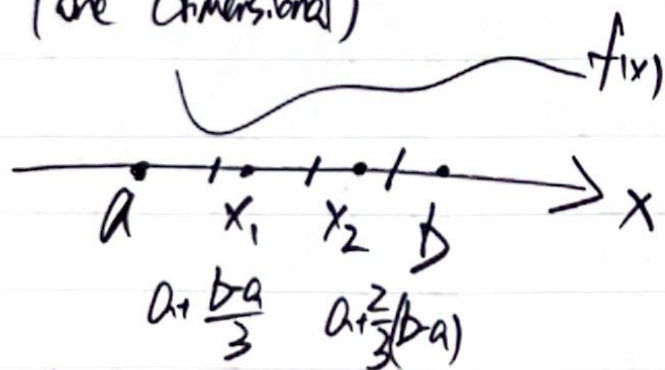
initial solution x_0

$$x_{k+1} = x_k + \alpha_k p_k$$

k : index of iteration

termination condition: $\|x_{k+1} - x_k\| < \epsilon$ / $|f(x_{k+1}) - f(x_k)| < \epsilon$

* Line Search
 (one dimensional)



$$\text{OR: } \begin{cases} b - \frac{F_{n-1}}{F_n}(b-a) = x_1 \\ a + \frac{F_{n-1}}{F_n}(b-a) = x_2 \end{cases}$$

F : Fibonacci series

1. 1. 2. 3. 5. 8. ...
 start from (a, b)

$$x^* = \frac{x_1 + x_2}{2}$$

if $f(x_1) > f(x_2)$ or $f(x_1) < f(x_2)$ or $f(x_1) = f(x_2)$
 repeat $(a, b) \sim (a, b)$ $\sim (a, b)$
 $\downarrow$ $\downarrow$ $\downarrow$
 x_1 x_2 x_1, x_2

"bigger one moves inside!"

1. Gradient-based method for NLP.

1° Steepest descent method.

$$X_{k+1} = X_k - \alpha_k \nabla f(X_k)$$

$$P_k = -\nabla f(X_k)$$

from X_0
set $\nabla f(x)$

$$\text{Substitute } X_k \Rightarrow X_{k+1} = X_k - \alpha_k \nabla f(X_k)$$

$$f(X_{k+1}) = f(X_k - \alpha_k \nabla f(X_k))$$

$$0 = \nabla_{\alpha} f(w) \Rightarrow \alpha_k$$

$$X_{k+1} = X_k - \alpha_k \nabla f(X_k)$$

2° Newton's Method Second-order Taylor series:

$$\tilde{f}(x) = f(X_k) + \nabla f(X_k)^T (x - X_k) + \frac{1}{2} (x - X_k)^T \nabla^2 f(X_k) (x - X_k)$$

$$\nabla \tilde{f}(x) = 0 + \nabla f(X_k) + \nabla^2 f(X_k) (x - X_k) = 0$$

$$x = X_k - [\nabla^2 f(X_k)]^{-1} \nabla f(X_k)$$

$$\therefore X_{k+1} = X_k - \alpha_k [\nabla^2 f(X_k)]^{-1} \nabla f(X_k)$$

3° Gauss-Newton method

Suppose objective function is quadratic:

$$\min \frac{1}{2} \|f(x)\|^2 \Rightarrow \frac{1}{2} f(x)^T f(x) \quad f(x) = \begin{bmatrix} f_1(x) \\ \vdots \\ f_m(x) \end{bmatrix}$$

first order Taylor series: $f(x, X_k) = f(X_k) + \nabla f(X_k)^T (x - X_k)$

Substitute

$$\nabla f(X_k) = [\nabla f_1(X_k) \dots \nabla f_m(X_k)]$$

$$= \frac{1}{2} [f(x) + \nabla f(X_k)^T (x - X_k)]^T [f(x) + \nabla f(X_k)^T (x - X_k)]$$

$$= \nabla f(X_k)^T f(X_k) + \nabla f(X_k) \nabla f(X_k)^T (x - X_k) = 0$$

$$X_{k+1} = X_k - \alpha_k [\nabla f(X_k) \nabla f(X_k)^T]^{-1} \nabla f(X_k)^T f(X_k)$$

4. Conjugate Gradient Method

$$\min f(x) = \frac{1}{2} x^T Q x - b^T x$$

step 1 let $p_0 = -\nabla f(x_0)$

step 2: $p_k = -\nabla f(x_k) + \sum_{i=1}^k \frac{\nabla f(x_k)^T Q p_i}{p_i^T Q p_i} p_i$ iteration

step 3: $\alpha_k = \arg \min f(x_k + \alpha p_k)$

$$\nabla f(x_k + \alpha p_k) = 0 \Rightarrow \alpha = ?$$

$$\begin{cases} p_i^T Q p_j = 0 \\ Q = W^T W \end{cases} \Rightarrow \begin{cases} p_i^T W^T W p_j = 0 \\ \zeta_i = W p_i \end{cases}$$

$$\therefore \zeta_i^T \zeta_j = 0$$

eg: $\min_{x_1, x_2} \frac{x_1^2}{16} + \frac{x_2^2}{4}$
 given $x_0 = (1, 1)^T$
 $\nabla f(x) = \begin{bmatrix} \frac{x_1}{8} \\ \frac{x_2}{2} \end{bmatrix}$ $Q = \begin{bmatrix} \frac{1}{8} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$

$$p_0 = -\nabla f(x_0) = \begin{bmatrix} -\frac{1}{8} \\ -\frac{1}{2} \end{bmatrix}$$

$$\alpha_k = \frac{-\nabla f(x_k)^T p_k}{p_k^T A p_k}$$

$$p_k^T A p_k$$

Hessian Matrix: $\begin{bmatrix} \frac{1}{8} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$

$$\alpha_0 = \frac{\begin{bmatrix} -\frac{1}{8} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} -\frac{1}{8} \\ -\frac{1}{2} \end{bmatrix}}{\begin{bmatrix} -\frac{1}{8} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{8} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -\frac{1}{8} \\ -\frac{1}{2} \end{bmatrix}} = 2.923$$

$$x_1 = x_0 + \alpha_0 p_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2.923 \begin{bmatrix} -\frac{1}{8} \\ -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 0.7 \\ -0.04 \end{bmatrix}$$

Repeat $\nabla f(x_k) = \begin{bmatrix} \frac{0.7}{8} \\ \frac{-0.04}{2} \end{bmatrix} = \begin{bmatrix} 0.0875 \\ -0.02 \end{bmatrix}$

$$\beta_k = \frac{\nabla f(x_k)^T A p_k}{p_k^T A p_k}$$

$$* \beta_0 = \frac{\nabla f(x_0)^T A p_0}{p_0^T A p_0}$$

↓

$$p_1 = -\nabla f(x_0) + \beta_0 p_0$$

$$\alpha_1 = \frac{-\nabla f(x_1)^T p_1}{p_1^T A p_1}$$

$$x_2 = x_1 + \alpha_1 p_1$$

$$\beta_1 = \frac{\nabla f(x_2)^T A p_1}{p_1^T A p_1}$$

$$p_2 = -\nabla f(x_2) + \beta_1 p_1$$

$$\alpha_2 = \frac{-\nabla f(x_2)^T p_2}{p_2^T A p_2}$$

$$x_3 = x_2 + \alpha_2 p_2$$

⋮

repeating

Direct Search Method

10. Coordinate descent method

e.g: $\min f(x) = x_1^2 + x_2^2 - x_1 x_2 - 10x_1 - 4x_2 + 60$

Start with $x_1 = 0$ $f(x) = x_2^2 - 4x_2 + 60$ $x_2 = 2$

min: $\frac{-b}{2a} = \frac{4}{2} = 2$ $x_2 = 2$

With $x_2 = 2$ $f(x) = x_1^2 - 12x_1 + 56$, $x_1 = 6$

$x_1 = 6$ $f(x) = \dots$

11. Exterior & Interior Point Method

$\min f(x)$

s.t. $g_j(x) \leq 0$, $j = 1, \dots, m$

$P(x, M) = f(x) + M \sum_{i=1}^m \left\{ \max[0, g_i(x)] \right\}^2$
 $M \rightarrow \infty$

$g_i(x) \leq 0$

$\phi(x, r) = f(x) - r \sum_{i=1}^m \frac{1}{g_i(x)}$

$r \rightarrow 0$

OR

$= f(x) - r \sum_{i=1}^m \ln(-g_i(x))$

Exterior-Point Method ex: $\max f(x) = x_1 x_2$

$$\text{s.t. } g_1(x) = x_1^2 - x_2 \leq 0$$

$$g_2(x) = -x_1 \leq 0$$

$$P(x, M) = x_1 + x_2 + M \left\{ \max[0, x_1^2 - x_2] \right\}^2 + M \left\{ \max[0, -x_1] \right\}^2$$

$$\frac{\partial P}{\partial x_1} = 1 + 2M \underset{\max}{(x_1^2 - x_2)} \cdot 2x_1 + 2M \underset{(-1)}{\max[0, -x_1]} \stackrel{\parallel}{\underset{(-1)}{\sum_{i=1}^{m+2}}} = 0!$$

$$\left\{ \frac{\partial P}{\partial x_1} = 1 + 2M \max[0, x_1^2 - x_2] (-1) = 0 \right.$$

"max" $\Downarrow$

$$\left\{ \begin{aligned} 1 + 4M x_1 (x_1^2 - x_2) + 2M x_1 &= 0 \\ 1 &= 2M (x_1^2 - x_2) \end{aligned} \right. \Rightarrow \begin{cases} x_1 = \frac{1}{2(M+1)} \\ x_2 = \frac{1}{4(M+1)^2} - \frac{1}{2M} \end{cases}$$

if $M \rightarrow \infty$ then $\left\{ \begin{aligned} x_1 &= 0 \\ x_2 &= 0 \end{aligned} \right.$ Optimal solution

Interior-Point Method: ex. $\min f(x) = \frac{1}{3}(x_1+1)^3 + x_2$

$$\text{s.t. } g_1(x) = x_1 - 1 \geq 0 \Rightarrow 1 - x_1 \leq 0$$

$$g_2(x) = x_2 \geq 0 \Rightarrow -x_2 \leq 0$$

$$\therefore Q(x, r) = \frac{1}{3}(x_1+1)^3 + x_2 - r \left(\frac{1}{1-x_1} + \frac{1}{-x_2} \right)$$

$$\left\{ \begin{aligned} \frac{\partial Q}{\partial x_1} &= 0 = (x_1+1)^2 - \frac{r}{(x_1-1)^2} \\ \frac{\partial Q}{\partial x_2} &= 0 = 1 - \frac{r}{x_2^2} = 0 \end{aligned} \right.$$

$$+ \frac{r}{x_1-1} + \frac{r}{x_2}$$

$$\Rightarrow \left\{ \begin{aligned} x_1 &= \sqrt{1+\sqrt{r}} \\ x_2 &= \sqrt{r} \end{aligned} \right.$$

if $r \rightarrow 0$ then $\begin{cases} x_1 = 1 \\ x_2 = 0 \end{cases}$

e.g. $\min (x_1 - 1)^2 + (x_2 - 1)^2$

s.t. $x_1 + x_2 \leq 1$

1° Use KKT-condition

$$f(x) = (x_1 - 1)^2 + (x_2 - 1)^2$$

$$g(x) = x_1 + x_2 - 1 \leq 0$$

① $\nabla f + \lambda \nabla g(x) = 0$

$$= [2(x_1 - 1) + \lambda, 2(x_2 - 1) + \lambda] = 0$$

$$\therefore x_1 = x_2 = 1 - \frac{\lambda}{2}$$

② $\lambda g(x) = 0 \quad g(x) = 2 - \lambda - 1 = 1 - \lambda \quad \therefore \lambda(1 - \lambda) = 0$
 $\therefore \lambda = 0, 1$

if $\lambda = 0$ $x_1 = 1, x_2 = 1$ infeasible

if $\lambda = 1$ $x_1 = \frac{1}{2} = x_2$ ✓ feasible $\rightarrow f = z^* = \frac{1}{2}$

2° Use Interior Point Method.

$$Q(x, r) = (x_1 - 1)^2 + (x_2 - 1)^2 - \frac{r}{x_1 + x_2 - 1} \ln(1 - x_1 - x_2)$$

$$\therefore \frac{\partial Q}{\partial x_1} = 2(x_1 - 1) + \frac{r}{1 - x_1 - x_2} = 0 \Rightarrow x_1 = x_2$$

$$\frac{\partial Q}{\partial x_2} = 2(x_2 - 1) + \frac{r}{1 - x_1 - x_2} = 0 \quad x = f(r) \dots \quad r \rightarrow \infty$$

always check if it's feasible!

especially Interior > Exterior w/ dual need

12. Dynamic Programming:

Bellman-equation:

$$J_t(x_t) = \max_{u_t \in U(x_t)} \left\{ g(x_t, u_t) + J_{t+1}(x_{t+1}) \right\}, t=1 \dots T$$

In a map.

J : cost total

g : road cost

$U \& u$: road

x : point/node

e.g. max $Z = \underbrace{2x_1}_{g_1} \underbrace{x_2^2}_{g_2} \underbrace{x_3}_{g_3}$

s.t. $x_1 + x_2 + x_3 = C$

$x_1, x_2, x_3 \geq 0$

$$S_0 = 0 \xrightarrow{x_1} 0 \xrightarrow{x_2} 0 \xrightarrow{x_3} S_3 = S_2 + x_3$$

$S_1 = S_0 + x_1$ $S_2 = S_1 + x_2$

① $J_3(S_3) = 1$

② $J_2(S_2) = \max_{0 \leq x_2 \leq C - S_2} \{ g_2(x_2) \cdot J_3(S_3) \} = C - S_2$

③ $J_1(S_1) = \max_{0 \leq x_1 \leq S_2 - S_1} \{ g_1(x_1) \cdot J_2(S_2) \} = \max_{0 \leq x_1 \leq S_2 - S_1} \{ 2x_1^2 (C - S_2) \}$

$= 2x_1^2 (C - S_1 - x_1)$

to find max: $f' = 0$

